Lab 3: Architectural Style Selection & Justification

Food Bank Surplus Redistribution Platform

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I chose Layered Architecture for the Food Bank Surplus Redistribution Platform.

Architectural Choice

The system is organized into three horizontal layers: a Presentation Layer (User Interface Component), a Business Layer (Food Donation Management, Reservation Management, and Notification Service Components), and a Data Layer (Food Bank Database Component). Each layer has a clear responsibility and communicates through defined interfaces, keeping user interaction, business processing, and data storage separated.

Two Reasons for This Choice

1. **Separation of donation and reservation responsibilities:** The platform has distinct operations for posting surplus food, managing reservations, and notifying nearby shelters. Layered architecture keeps these responsibilities within the Business Layer, making the system easier to understand and maintain as features are added.
2. **Centralized and consistent food information:** Food batches, quantities, expiry information, dietary tags, and reservations need to remain consistent. A single Data Layer provides centralized storage and avoids unnecessary synchronization between separate data stores. This is appropriate for an institution-scale platform without the operational complexity of Microservices.

Security Advantage

All access to the Food Bank Database is controlled through the Business Layer rather than allowing the User Interface to communicate with the database directly. Validation and authorization can therefore be applied before food listings or reservation records are created, updated, or retrieved. This helps protect donor, shelter, food-batch, and reservation information.

Performance Benefit

The Notification Service can process alerts for high-quantity perishable food separately from normal user interactions. The layered structure also avoids the additional network hops and serialization overhead that a Microservices architecture could introduce for every internal request. This supports responsive food listing and reservation operations for the expected platform load.

DIAGRAM

